DO NOW — RECALL

# Paging, segmentation and virtual memory

Name: _____ Class: _____ Date: _____

**1. What is RAM used for?**

**2. What are remote virtual storage files?**

**3. Why might a process have to wait on memory I/O?**

**4. A 4-row, 3-column zero-indexed array: give its final valid index pair and number of values.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Active program and data.

**2. Model answer**

Files hosted on network-accessible servers.

**3. Model answer**

Required page/data is not resident and must be loaded.

**4. Model answer**

Index (3,2), twelve values.

**Vocabulary — know these terms**

page — Virtual fixed-size memory block

frame — Physical fixed-size RAM block

page fault — Page absent from RAM

segmentation — Logical variable-length regions

NEW LEARNING

# Read, annotate and explain

### Understand

Pages and frames have equal fixed sizes; the page table maps virtual pages to physical frames.

### Trace

Divide the virtual address by page size for the page number; use remainder as the offset.

### Explain

A page fault requires a non-resident page to be loaded. Variable-sized segmentation has different fragmentation characteristics.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

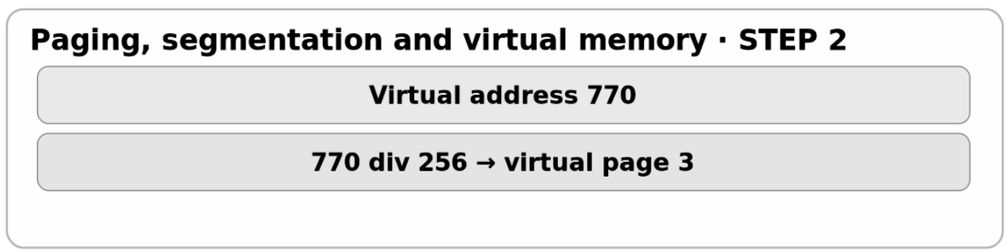
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**



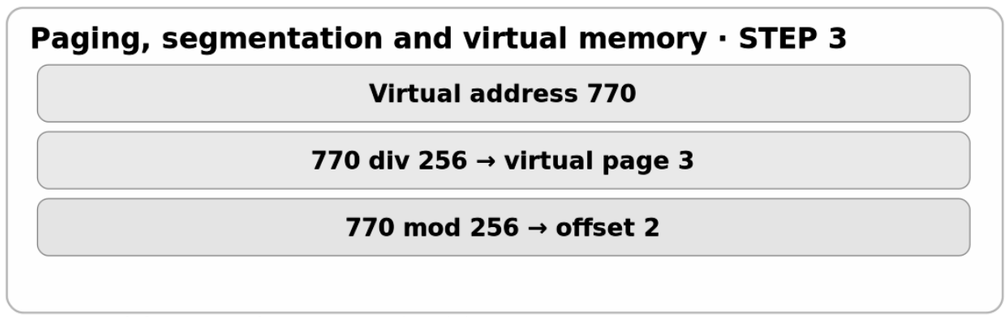

Paging, segmentation and virtual memory · STEP 1

**New Learning**




Paging, segmentation and virtual memory · STEP 2

**New Learning**




Paging, segmentation and virtual memory · STEP 3



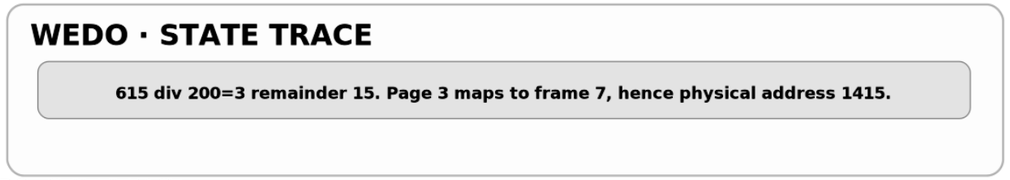


# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Virtual address 615 with page size 200, page 3 → frame 7. Calculate physical address.**

615 div 200=3 remainder 15. Page 3 maps to frame 7, hence physical address 1415.




WEDO · STATE TRACE

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

### Guided 1 — CHECK

Predict first
615 div 200=3 remainder 15. Page 3 maps to frame 7, hence physical address 1415.

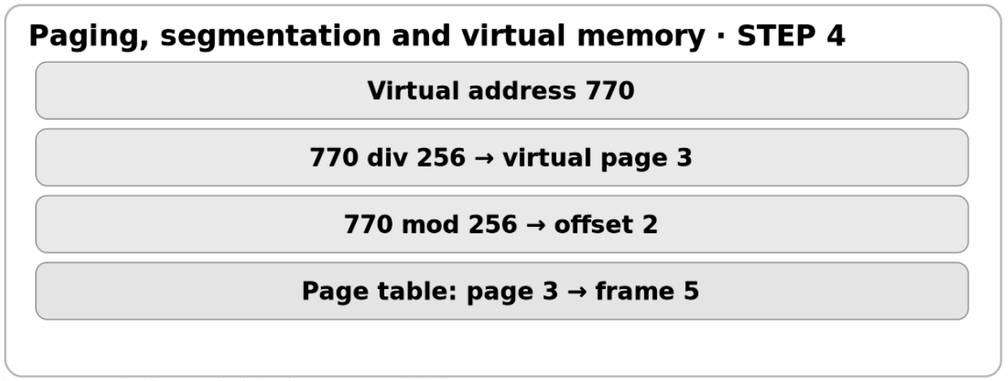




# New Learning · original lesson diagrams

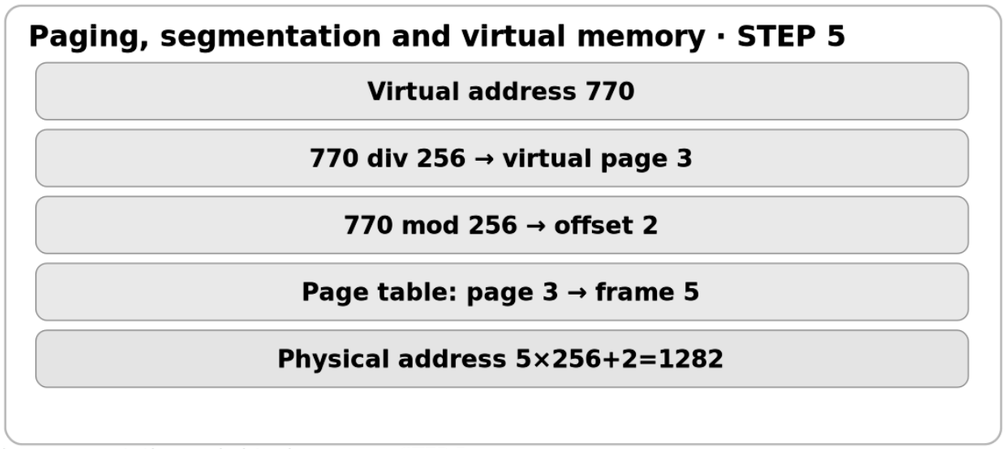
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Paging, segmentation and virtual memory · STEP 4

**New Learning**




Paging, segmentation and virtual memory · STEP 5

I DO — WORKED EXAMPLE

# Follow the worked method

**Translate virtual address 520 with page size 128 and page 4 mapped to frame 9.**

**Predict first**

520 div 128=4 remainder 8; frame 9 base=1152, plus 8 gives 1160.





EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**Paging changes the offset within the page when it changes frame.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

**GREEN PEN — Correct and explain**

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

**Question 3**

Compare paging and segmentation and explain thrashing. [6]



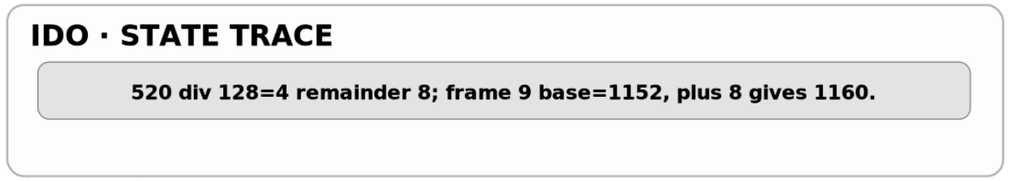


# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Translate virtual address 520 with page size 128 and page 4 mapped to frame 9.**

520 div 128=4 remainder 8; frame 9 base=1152, plus 8 gives 1160.




IDO · STATE TRACE

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

Virtual address 615 with page size 200, page 3 → frame 7. Calculate physical address.
Source code / input:
page=3; offset=15; physical=7×200+15=1415





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Define page, frame, page table and page fault. [4]

### Question 2

Page size 256, VA 770, page 3 mapped to frame 5. Find physical address. [6]